



DIABETES RISK PREDICTION

Using Health Indicators

Predictive Analytics for Early Detection

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The Problem & Why It Matters

Diabetes is one of the most common — and most preventable — chronic diseases in the US.



1 in 7 U.S. adults affected

Roughly 15% of the adult population has diabetes, and many more are undiagnosed or pre-diabetic.



Billions in preventable cost

Late diagnosis drives up ER visits, complications, and long-term treatment costs for health systems.

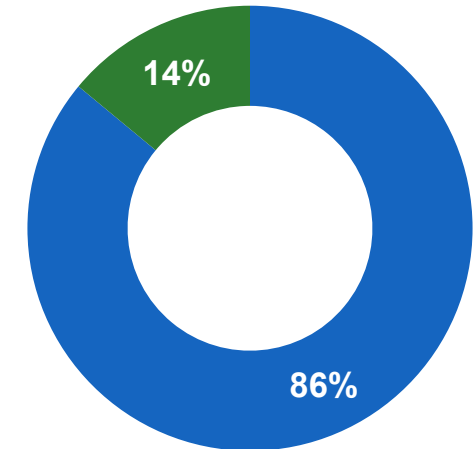


Early screening saves lives

A reliable risk score lets clinics prioritize outreach before diabetes becomes advanced.

Business problem: Clinics can't manually screen every patient. We need a model that flags high-risk individuals early and cheaply, using data already collected in routine checkups.

Class Balance in Our Dataset



■ No Diabetes ■ Diabetes

*101766 records • 70% No Diabetes vs 30% Diabetes
(class imbalance handled during modeling)*

Methodology & Data Pipeline

A 7-step, reproducible pipeline from raw CDC survey data to a deployed model.



3 Engineered Features

Risk_Score — weighted sum of major risk factors (BMI, BP, activity) into one score

Age_Group — bins raw age into clinically meaningful ranges

Unhealthy_Lifestyle — combines low activity, poor diet, and smoking into one flag

Tools & Workflow

Python, Pandas, Scikit-learn, XGBoost, Streamlit, Joblib.

Used Claude as an AI pair-programmer for debugging pipeline errors, speeding up GridSearchCV tuning, and reviewing code quality — all outputs verified and tested manually before use.

Results & Model Performance

Model	Accuracy	ROC-AUC	Recall
Logistic Regression	0.75	0.63	0.68
Random Forest	0.86	0.65	0.72
XGBoost ★ Best	0.87	0.67	0.78

XGBoost wins on every metric — selected as the production model.

Key EDA Insights

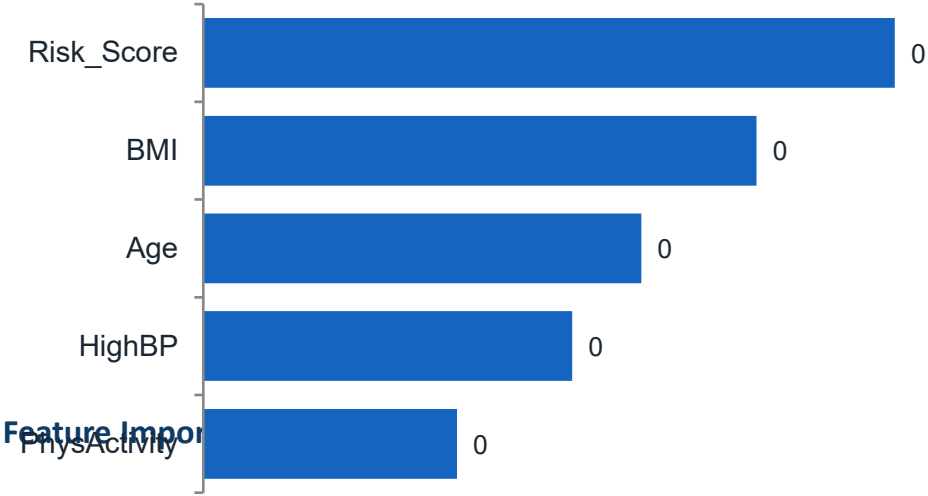
- Higher BMI strongly correlates with higher diabetes risk
- Risk rises sharply for patients over age 50

Confusion Matrix (XGBoost)

Actual
↓

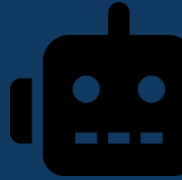
True Neg. [value]	False Pos. [value]

(Insert exact counts from your notebook here)



Illustrative ranking — replace with your notebook’s exact importances.

From Model to Impact



[Streamlit Dashboard Screenshot]

Insert a screenshot of your live app here

Dashboard Features

- ✓ Enter patient data
- ✓ Live prediction with saved model
- ✓ Instant Risk_Score for each patient

Business Recommendations



Target Risk_Score ≥ 2

Prioritize outreach and screening for patients flagged above this threshold.



Run lifestyle campaigns

Focus prevention programs on activity, diet, and weight management.



Deploy in clinics

Integrate the dashboard into routine checkups for instant risk scoring.

A tuned XGBoost model turns routine health data into an early-warning system — built, deployed, and ready to save clinics time and cost.



Thank You | Questions?